Differences in the Behavioral Phenotype of ASD in a Population Sample of Somali, White, Non-Somali Black, and Hispanic Children

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INTRODUCTION

The Minneapolis Somali Autism Spectrum Disorder Prevalence Project (MSASDPP) was conducted to determine if more Somali children had autism spectrum disorder (ASD) than non-Somali children. In addition to estimating ASD prevalence, the project identified differences in characteristics of ASD across racial and ethnic groups. The overall project found that white and Somali children did not differ statistically in ASD prevalence; black (non-Somali) and Hispanic had significantly lower rates than white and Somali children.

Race and ethnicity	Children with ASD identified/ Total population	Prevalence estimate (prevalence per 1,000 children)	95% Confidence interval*
Overall	255 of 12,329	**1 in 48** (20.7 per 1,000)	18.3 – 23.4 per 1,000
Somali	31 of 1,007	**1 in 32** (30.8 per 1,000)	21.6 – 43.8 per 1,000
White	120 of 4,336	**1 in 36** (27.7 per 1,000)	23.1 – 33.1 per 1,000
Black (non-Somali)	53 of 3,312	**1 in 62** (16 per 1,000)	12.2 – 20.9 per 1,000
Hispanic	30 of 2,399	**1 in 80** (12.5 per 1,000)	8.7 – 17.9 per 1,000

Note: We are unable to report on Asian/Pacific Islanders and Native Americans due to their low numbers.
* 95% confidence interval is the range in which an estimate is likely to fall.



RESULTS

Somali children with ASD were far more likely to have intellectual disability than children with ASD in all other racial and ethnic groups.

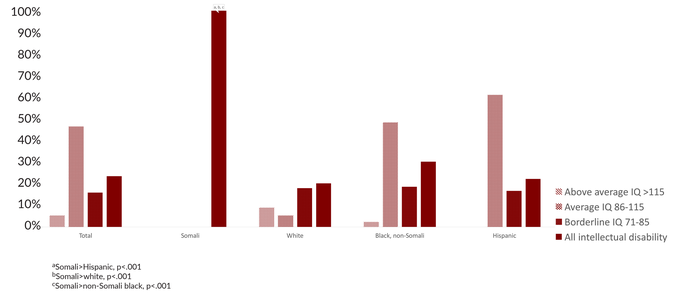
DSM-IV Social Interaction: Somali children were more likely than white and Hispanic children to have deficits in seeking to share enjoyment.

DSM-IV Communication: Somali children were more likely than all other racial/ethnic groups to have lack of pretend play documented in their records.

DSM-IV Restricted, Repetitive behaviors: Somali children were more likely than Hispanic children to have preoccupations with parts of objects and more likely than Hispanic children to have stereotyped and repetitive motor mannerisms. Somali and White children had a greater number of DSM-IV repetitive behavior symptoms documented in their records than Hispanic children, and Somali children were more likely to have co-occurring concerns of abnormalities in eating and drinking patterns than White and non-Somali Black children.

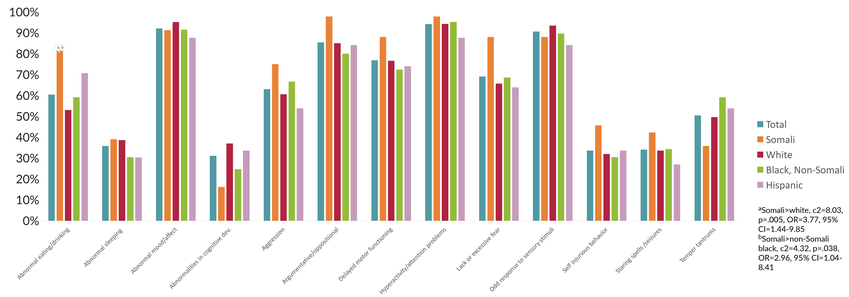
Associated Features: Somali children were significantly more likely to have atypical eating/drinking habits than white and black non-Somali children.

Cognitive Levels for Confirmed ASD Cases with Cognitive Data in Their Records



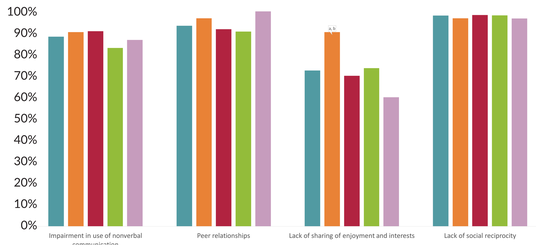
[a]Somali>Hispanic, p<.001
[b]Somali>white, p<.001
[c]Somali>non-Somali black, p<.001

Associated Features: All ASD Cases

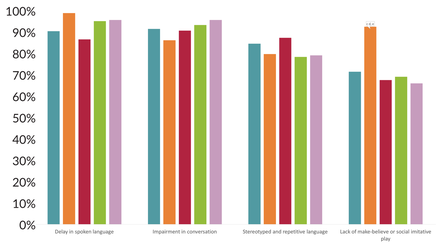


[a]Somali>white, c2=8.03, p=.005, OR=3.77, 95% CI=1.44-9.85
[b]Somali>non-Somali black, c2=4.32, p=.038, OR=2.96, 95% CI=1.04-8.41

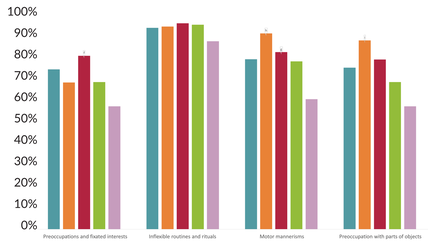
Social Interaction

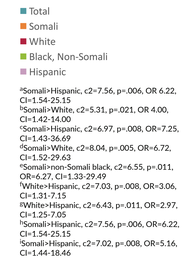


Communication



Restrictive, Repetitive Behavior





[a]Somali>Hispanic, c2=7.56, p=.006, OR 6.22, CI=1.54-25.15
[b]Somali>White, c2=5.31, p=.021, OR 4.00, CI=1.42-14.00
[c]Somali>Hispanic, c2=6.97, p=.008, OR=7.25, CI=1.43-36.69
[d]Somali>White, c2=8.04, p=.005, OR=6.72, CI=1.52-29.63
[e]Somali>non-Somali black, c2=6.55, p=.011, OR=6.27, CI=1.33-29.49
[f]White>Hispanic, c2=7.03, p=.008, OR=3.06, CI=1.31-7.15
[g]White>Hispanic, c2=6.43, p=.011, OR=2.97, CI=1.25-7.05
[h]Somali>Hispanic, c2=7.56, p=.006, OR=6.22, CI=1.54-25.15
[i]Somali>Hispanic, c2=7.02, p=.008, OR=5.16, CI=1.44-18.46

METHODS

The project used a multi-step review of medical and educational records of children who were age 7 to 9 years and who had at least one parent living in Minneapolis in 2010. Data documenting DSM-IV-TR autism symptoms, intellectual ability, and related behavioral and medical concerns were abstracted from school special educational evaluations and from medical records of children who had specific DSM-IV diagnostic codes documented (see Rice et al. 2007 for details on methodology). Records for children were abstracted if they contained "triggers", including: (1) An ASD diagnosis/description, an ASD special education classification, or an ASD test/assessment, and (2) ASD behavioral triggers. ASD behavioral triggers are defined as social behaviors associated with a diagnosis of ASD. A review included all available records for children from birth until the end of the 2010 calendar year. Children with records at multiple sites, both education and clinic, had their data compiled into one composite record to be reviewed, ensuring no overlap. Abstracted files were de-identified and sent to trained clinician reviewers. Race/ethnicity information was not deidentified. Clinicians reviewed the abstracted data and determined ASD case status based on DSM-IV-TR criteria.

DISCUSSION

The project's main finding was that Somali children were far more likely to have ASD + intellectual disability (ASD+ID) documented in their records than other racial/ethnic groups. A few additional differences in DSM-IV symptom endorsement were found across groups.

An important caveat is that sample size was not adequate in most cases to compare differences in rates of symptom endorsement when controlling for presence of ID. When the sample was limited to only children with ASD+ID, high rates of DSM-IV symptoms and associated features were endorsed for all groups. This suggests differences in ASD-specific characteristics across groups may have been driven by level of cognitive impairment rather than by race/ethnicity.

IMPLICATIONS

The findings of lower rates of ASD among non-Somali Black and Hispanic children point to potential bias and disparities in diagnostic practices and access to specialized evaluations for ASD. Results also indicate that Somali children may require a higher level of service in terms of frequency, intensity, and coordination than other children with ASD due to their high rate of co-occurring intellectual disability.

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Rice, C., Baio, J., Van Naarden, Braun, K., Doernberg, N., Meaney, F., Kirby, R. (2007). ADDM Network. A public health collaboration for the surveillance of autism spectrum disorders. Paediatric Perinatal Epidemiology, 21, 179–190.

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